Null Controls Are Necessary but Not Sufficient: Auditing Acceptance Guards for Scientific Discovery and Interpretability

Project Aether https://github.com/Real-Project-Aether/project-aether
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Abstract

An automated-discovery system needs something that proposes candidates and something that decides which to keep. This paper audits the deciders, reporting three quantities for every acceptance guard: how often it accepts candidates built to carry nothing (NAR), how often it accepts known positives (PAR), and how often what it accepts survives an independent consequence test (CAR). All but one of the descriptive guards we audit, the checks each operation's definition suggests, accept candidates that carry nothing the operation is meant to find. A coarse-graining that maps every state to zero closes perfectly on 96% of 129 chaotic systems from a public benchmark, while a refutable second guard rejects it on all 129. For representational correspondence the cause is structural: linear CKA gives the same score to a genuine correspondence and to that map composed with a random rotation, and more data cannot separate them. Held out, the alignment guard still accepts 0.60 of the nulls (0.80 on ESM-2 protein models). A localised difference-patching consequence test refuses every null while accepting every genuine correspondence, and on ESM-2 it accepts both built-in positive controls (PAR $= 1.00$). Rejecting nulls is not sufficient. On sparse-autoencoder features the correlational guard accepts none of 16 null controls. Yet an independent ablation test supports CAR@150 $= 0.45$ of the 405 selected features it can test against matched controls, against 0.005 for those controls. Where that test's own positive control is valid, it detects 0.90 of directions chosen to be causal and supports 0.40 of the tested features. Code and data: https://real-project-aether.github.io/project-aether/.

1 Introduction

An automated-discovery system needs two things: something that proposes candidates, and something that decides which to keep. Deciders are receiving attention, but mostly as allocators of compute [Foster et al., 2026]; Posner et al. [2026] list verifying recovered structure, rather than relying on prediction accuracy alone, as an open challenge. This paper is about the decider that runs last, the one that certifies a candidate as a result. The acceptance tests that each operation's definition suggests turn out to accept candidates that carry none of the structure the operation looks for.

We built four discovery operations, chosen from a corpus of scientific prizes (Appendix G). Each takes a theory or a system and returns a modified one. For each we wrote the acceptance test its own definition suggests, the middle column of Table 1.

Each of the four accepts candidates that carry nothing the operation seeks. For L4 this means positing a new entity on an imbalance that a calibration error produces. In one instance the guard

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destroys the structure the operation was meant to find. None of the four failures is visible in a demonstration on cases whose answers are known.

Contributions.

1. **One audit protocol** (Sec. 2): null acceptance, positive acceptance and consequence agreement, with the candidate distribution stated, applied unchanged across mechanisms that share no machinery.
2. **The guard finding** (Secs. 3, 4): four discovery operations for physics, the acceptance test each definition suggests, and a failure in each. The L2 failure replicates on 129 chaotic systems from the `dysts` benchmark [Gilpin, 2021], with thresholds imported unchanged.
3. **The same audit in three interpretability settings** (Sec. 6): representational correspondence between language models and between **ESM-2 protein models**, and sparse-autoencoder features — which together show *two* distinct ways an acceptance test fails, only one of which the physics made visible.

Scope of the concern. The descriptive guards we audit are the checks an operation's definition suggests, and each is reported as evidence in practice. Closure is the defining property of a Markovian coarse-graining [Zwanzig, 1960], and it is the prediction term of a recent objective for discovered world models when prediction is taken in the learned coordinates [Posner et al., 2026]. Linear CKA is widely used to compare layers across networks [Kornblith et al., 2019, Raghu et al., 2021, Davari et al., 2023], and a high score is read as two layers representing the same thing. Sec. 6 reports one descriptive guard that rejects every null. Running a null is the cheapest way to learn whether a given check accepts empty inputs.

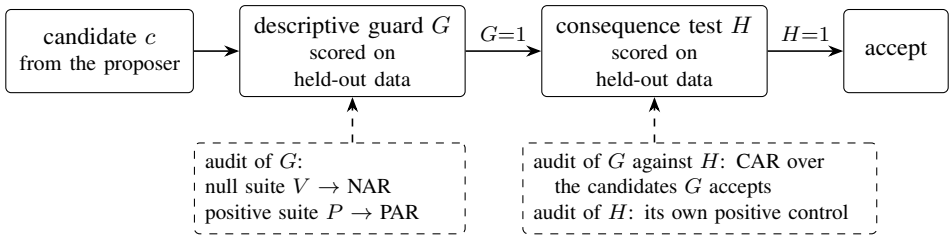


Figure 1: The audit. A guard is scored three ways: against a null suite V carrying nothing the operation seeks (NAR), a positive suite P carrying it by construction (PAR), and an independent consequence test on the candidates it accepts (CAR, over a stated distribution $\mathcal{C}$). The consequence test gets a positive control of its own. In the interpretability experiments and on the external benchmark, acceptance uses data the candidate was not fitted to, and in-sample scores appear only for contrast. X2 uses three disjoint splits (60/20/20: autoencoder training, guard, consequence test). X1 and X3 fit the map on half the data and score both tests on the other half. On the external benchmark the coarse law is fitted on trajectories other than the one it is scored on.

2 The guard-auditing framework

Fig. 1 states the protocol. Write $G(c; D) \in \{0, 1\}$ for a descriptive acceptance guard applied to a candidate c given data D, and let $V = \{c_1, \ldots, c_m\}$ be a *prespecified, finite* null suite: candidates carrying nothing the operation is meant to find. Of the three quantities, the first two concern nulls and answer different questions:

$$\widehat{\mathrm{NAR}}(G; V, D) = \frac{1}{m} \sum_{c \in V} G(c; D), \qquad \mathrm{AnyNullPass}(G; V, D) = \max_{c \in V} G(c; D). \tag{1}$$

The **null acceptance rate** says how often the tested nulls get through; **AnyNullPass** says whether any single null defeats the guard. V holds candidates of a null type. A row that perturbs the consequence test instead of the candidate is a control on H; it is reported beside the nulls and is not counted in V. Each NAR below is given with its count. Neither is a supremum over an optimised family: V is a list we write down, so these are estimates against that list.

Null acceptance is not the only way a guard fails. A guard may reject every null and still endorse candidates that a second, independent test does not support, so we also report the **consequence agreement rate**

$$\mathrm{CAR} = \Pr[H(c) = 1 \mid G(c) = 1,\ c \notin V]\,, \tag{2}$$

where H is a consequence or intervention test fixed before it is scored. H is an independent test that can disagree with G; CAR measures that disagreement and does not treat H as ground truth. In the experiments, guard 1 is a descriptive guard G and guard 2 a consequence test H. The two failure modes are separately diagnosable. High NAR is **null acceptance**; low CAR is **unsupported acceptance**. Sec. 6 shows a guard with an NAR of zero and a low CAR, so the second is not a corollary of the first.

CAR is conditional on $c \sim \mathcal{C}$, the distribution of candidates actually put to the consequence test, so a different proposer or selection policy gives a different number for the same guard. We state $\mathcal{C}$ wherever CAR appears. In Sec. 6 it is the top K candidates by guard score within each arm (one model and layer), so the quantity is $\mathrm{CAR@}K$.

Both quantities describe what a guard refuses, and a guard that refuses everything scores $\mathrm{NAR} = 0$ while being worthless. We therefore also report the **positive acceptance rate** over a prespecified suite P of candidates constructed to carry what the operation seeks:

$$\widehat{\mathrm{PAR}}(G; P, D) = \frac{1}{|P|} \sum_{c \in P} G(c; D). \tag{3}$$

$1 - \mathrm{NAR}$ and PAR are a guard's specificity and sensitivity, each measured against a list written down in advance.

This has a consequence for CAR that is easy to miss. CAR counts how often H fires among the candidates G accepts, so a low CAR admits two readings: G accepts candidates nothing supports, or H cannot detect support that is present. CAR alone does not separate them. The consequence test therefore needs a positive control of its own, $\mathrm{PAR}(H; P)$, the rate at which H fires on candidates built to be detectable. A measured CAR must be read against that ceiling rather than against 1. Two consequences are used below. A matched control keeps $\Delta\mathrm{CAR}$, the CAR of accepted candidates minus that of matched controls, interpretable even when $\mathrm{PAR}(H; P)$ is low, because the same insensitive H scores both groups. And a positive control is valid when its own manipulation has the effect it was built to have. This is checkable, and in two of X2's four arms it does not hold (Sec. 6).

3 Four operations

Each operation turns a move the corpus records into a computation, and each needs *two* conditions to accept a candidate. Full definitions, implementations and all controls are in Appendix A.

L1, L3 and L4 in brief. L1 reads the directions along which no prediction moves, structural non-identifiability [Raue et al., 2009], as a property of a theory, and must separate probe invariance, dynamical invariance, and the local-versus-uniform test that distinguishes a gauge redundancy from a global symmetry. L3 requires what crosses between domains to be a claim the target did not have and its own dynamics can refute. L4 guards on the noise scale rather than significance. The audit below turns on L2, whose guards the external replication uses unchanged.

L2, coarse-graining. Where L1 asks what *parameter* detail predictions do not need, this asks what *state* detail the future does not need. A reduction Q *closes* if the coarse variables predict their own future without the fine state, with a normalised error (Sec. 5) below 0.25. Closure can be tested with either of two laws. The *learned* law is a regression from coarse states to coarse velocities, whose targets are the fine law evaluated at the lifted state (the fine state the coarse one maps back to). The *derived* law projects the known fine law [Zwanzig, 1960]. If derived closes and learned does not, the failure was our fitter. Appendix A states which law carries the guard in each experiment. The second guard is that the reduction must *pay*:

$$\mathrm{Pay}(Q) \;=\; \mathrm{clip}_{[-1,1]}\left(1 - \frac{\sum_{\phi \in \mathcal{E}} \|P(Q^{\dagger}Q\phi) - P(\phi)\|_{\Sigma}^{2}}{\sum_{\phi \in \mathcal{E}} \|P(\phi) - \bar{P}\|_{\Sigma}^{2}}\right), \qquad \|v\|_{\Sigma}^{2} = v^{\top}\Sigma^{-1}v, \tag{4}$$

Table 1: Four operations, four intuitive acceptance tests, four failures. The second guard in each row hands the system a claim it can refute; the first only describes. Any: AnyNullPass.

	the intuitive guard	NAR	Any	null suite V (type)	what the consequence test buys
L1	remove what no probe sees	**1.00**	1	4 charge-carrying directions (STRUCT.-PRESERVING)	8.6×10^{-1} vs $\leq 7.6\times10^{-10}$
L2	the coarse variables close	**0.67**	1	constant, random, fast modes (DEGEN., RANDOM.)	empty cases score -0.54, -0.50
L3	the structures match	**0.33**	1	6 broken-symmetry targets (CORR.-BREAKING)	4.7×10^{10}
L4	the imbalance is significant	**1.00**	1	2 calibration offsets, nothing hidden (SUB-THRESHOLD)	silent at 16σ, 31σ

where P is a prespecified observable vector, $Q^\dagger$ the adjoint of the reduction, and $\bar{P}$ and Σ the mean and covariance of P over an *ensemble* $\mathcal{E}$ of states; along one trajectory the denominator nearly vanishes. Weighting by Σ^{-1} makes the score invariant under any invertible linear map of P, so one threshold carries between systems whose observables have different units. $\mathrm{Pay}(I) = 1$ exactly, the clip makes -1 a floor, and nothing in the score is fitted (Appendix A). A reduction pays when $\mathrm{Pay}(Q) > 0.5$. Candidates come from linear proposers or from a language model, which proposes and never decides (Appendices B and D).

4 Auditing the guards: null acceptance

Table 1 sets the four cases side by side; the rest of this section takes them in turn, and Sec. 7 reports what the operations recover when both guards are applied.

L2: the constant reduction closes by construction. If $Q(x) = q_0$ for every x, the coarse trajectory is constant and the coarse law $\dot{q} = 0$ reproduces it exactly, so the closure error is 0 for every system, trajectory and horizon. **Any acceptance test resting on closure alone accepts a reduction to nothing.** It is also the most parsimonious candidate, so an objective that scores simplicity and self-prediction, the form of the mechanistic world-model objective of Posner et al. [2026] when prediction is taken in the learned coordinates, ranks it first (Appendix A). A reduction onto the *fast* modes closes for a related reason: fast modes decay to zero, so it describes nothing and evolves correctly. The experiments below measure what a criterion of this form accepts without a second condition, and what adding one costs. Of three vacuity controls, two close perfectly and only the pay guard of Eq. 4 refuses them, at -0.54 and -0.50 against a threshold of 0.50; Table 5 in Appendix A gives all three.

L1, L3 and L4. For L3, a target domain built *without* the source's structure passes the structural match in two seeds of six on held-out states, and in five of six in-sample. The transferred-claim test separates them: the weakest genuine transfer scores 4.7×10^{10} times better than the strongest control, a gap that reflects near-zero residuals on the real cases. For L1, five directions on the lattice are invisible to every probe; quotienting all five, as "remove what no measurement can see" instructs, eliminates one redundancy and **destroys four conservation laws**, each conserved to 7.6×10^{-10}. For L4, miscalibrating the measured output by 2.5% or 5%, with nothing hidden, gives an imbalance of 16σ or 31σ. A significance guard posits a carrier on both; the operation's noise-scale guard stays silent.

Our own wrong guards. Three earlier pay guards failed the check that a reduction keeping everything scores exactly 1.000, and an early X1 causal guard ranked a null above the real case (Table 8). None of the four failures above shows on cases with known answers, because there both guards agree.

5 External replication: the guards on 129 systems we did not build

Everything in Sec. 4 was measured on systems we built. That is the obvious objection to it, so we tested the L2 guards on the `dysts` benchmark of chaotic dynamical systems [Gilpin, 2021]: 137

published systems, assembled by someone else for a different purpose, with equations we did not choose.

Protocol. The closure test, the pay score of Eq. 4 and both thresholds are *imported* from the module used for Sec. 4, not reimplemented and not retuned. An earlier version of this work got two details wrong. The coarse law is now fitted on trajectories *excluding* the one it is evaluated on. Acceptance now rests on the **learned** law alone, where it used the better of the learned and the derived law. Neither correction changed the null result below. Both changed the positive one: under the superseded rule the real candidate closed on one system, and under the learned-only rule, which every benchmark number here uses, it closes on 0. 129 of 137 systems ran (Appendix H).

The empty reduction closes on 96% of a public benchmark. Closure error is the distance between the integrated coarse trajectory and the true one, divided by the true coarse trajectory's spread, with the divisor floored at 10^{-12}. Mapping every state to zero makes the distance exactly zero, so the error is 0.000 whatever the floor, on 124 of 129 systems. On the other 5 the error cannot be computed: the constant reduction lifts its coarse state to the fine-state origin, and there the published right-hand side is undefined (a division by zero or a NaN). A non-finite error counts as not closing. The pay score for the constant reduction never exceeds -0.041, far below the 0.50 threshold, so guard 2 refuses it on **129 of 129**. The thresholds were fixed on a different system before this benchmark was considered. Table 15 and Fig. 3 in Appendix H give the full result.

Coordinate dependence. With the origin moved well off the attractor, the lifted constant reduction closes on 126 of 129 systems, every system closes in at least one of the two coordinate systems, and where both errors are finite (121 systems) they are identical. A zero-dimensional reduction with no lift closes on all 129 (Appendix H).

Sensitivity, on a suite with ground truth. The dysts systems carry no ground truth, so we built 24 slow–fast systems whose slow subspace is the true coarse description by construction (Table 14, Appendix H). Closure here uses the derived law, which has nothing fitted. Handed the true subspace the guard pair accepts every system ($\mathrm{PAR} = 1.00$); handed the proposer's estimate, 0.71. Closure alone accepts 47 of 72 empty candidates ($\mathrm{NAR} = 0.65$), the pair none.

The real candidate. The slow-mode reduction closes on 0 of 129 systems. A linear projection onto slow eigen-directions of a mean Jacobian is the wrong object for a chaotic attractor, whose invariant manifold is curved, so on this benchmark the operation never fires, and the limit lies in our proposer.

6 Auditing the guards: three interpretability settings

Everything above is physics. The failure it describes is a property of acceptance criteria, so it should also appear in methods built for an unrelated purpose. We ran the same pair of guards on the two objects interpretability works with directly: a claimed correspondence between two models' representations, and a claimed meaning for a sparse-autoencoder feature. X1 and X2 use Pythia models [Biderman et al., 2023], X3 uses ESM-2, and all are scored on held-out data; details are in Appendix F.

X1, representational correspondence. A linear map W is fitted from one model's residual stream to another's, and the claim is that a direction in the first means what its image means in the second. Guard 1 is the alignment quality reported for such claims: held-out R^2 of the map, and linear CKA [Kornblith et al., 2019], accepting when either exceeds 0.5. Guard 2 is a *localised difference patch*: perturb model A's residual stream at one position, carry the same perturbation through W into B, and compare the induced changes in the two models' next-token distributions. Adding the mapped *difference* keeps B in distribution, and the causal test does not inherit the reconstruction quality that guard 1 measures.

Every candidate is a claim about a specific map, and is scored with the map it claims. The nulls are chosen to break the correspondence while preserving what guard 1 looks at: pairings permuted before fitting, the same map composed with a random rotation, a covariance-matched Gaussian target, a map fitted on disjoint text, and a randomly initialised target model. A sixth row permutes which donor B

receives while holding the map fixed. It is a control on the consequence test and is not counted in NAR. Across 4 model-layer configurations and 1920 interventions, Table 2 gives the result.

Table 2: X1 and X3, medians across configurations. Guard 1 accepts when R^2 or CKA exceeds 0.5. Causal: median correlation between the two models' induced shifts; guard 2 accepts above 0.2. AUROC: separation of a candidate's per-intervention scores from the genuine pair's. Accepted: both guards accept. **All nulls except the randomly initialised target, and also the donor-permuted control, have the same CKA as the genuine correspondence** to three decimals, because CKA scores the two sets of representations and never looks at the claimed map. Rows are generated from the result files.

candidate	type	R^2	CKA	guard 1	causal	AUROC	accepted
X1: Pythia pairs, 4 *configurations,* 480 *interventions each*							
the correspondence under test	real	0.82	0.61	4/4	+0.80	–	4/4
pairings permuted	corr.-breaking	-0.04	0.61	3/4	-0.00	0.995	0/4
the same map, randomly rotated	corr.-breaking	-0.96	0.61	3/4	-0.01	0.996	0/4
covariance-matched Gaussian target	randomised	-0.04	0.61	3/4	+0.00	0.995	0/4
fitted on disjoint text	task-confounded	0.65	0.61	3/4	+0.01	0.996	0/4
randomly initialised target	degenerate	0.19	0.03	0/4	-0.00	1.000	0/4
donor permuted at patch time	test control	0.82	0.61	4/4	-0.09	0.965	0/4
X3: ESM-2 pairs, 3 *configurations,* 288 *interventions each*							
identity map (code check)	positive	1.00	1.00	3/3	+1.00	–	3/3
adjacent layer, same model	positive	0.73	0.76	3/3	+1.00	–	3/3
the correspondence under test	real	0.47	0.65	3/3	+1.00	–	3/3
pairings permuted	corr.-breaking	-0.02	0.65	3/3	+0.01	1.000	0/3
residues permuted within protein	corr.-breaking	0.12	0.65	3/3	+0.02	1.000	0/3
the same map, randomly rotated	corr.-breaking	-0.64	0.65	3/3	+0.01	1.000	0/3
covariance-matched Gaussian target	randomised	-0.02	0.65	3/3	+0.01	1.000	0/3
randomly initialised target	degenerate	0.39	0.21	0/3	+0.03	1.000	0/3
donor permuted at patch time	test control	0.47	0.65	3/3	-0.16	0.974	0/3

More data helps the R^2 half of guard 1. Scored in-sample on eight stimuli, R^2 accepts 0.80 of the nulls (16 of 20: 5 nulls in 4 configurations), and at 8,192 stimuli 0.20. Held out on every position, guard 1 still accepts 0.60 (12 of 20), and the acceptances come from CKA. CKA is computed between two sets of representations, so it is unchanged by *any* claim about how they correspond. Rotating the map by a random orthogonal matrix destroys R^2 (to -0.96) and leaves CKA exactly where it was, and a guard that accepts on either criterion therefore accepts the rotation. Guard 2 refuses every null in every configuration ($\mathrm{NAR} = 0.00$) while accepting the genuine pair in all 4. Taken separately, held-out R^2 alone accepts 2 of 20 null evaluations (the map fitted on disjoint text) and the genuine pair in all 4 configurations; CKA alone accepts 12 nulls and rejects the genuine pair in 1. On ESM-2, R^2 alone accepts 0 of 15 nulls but rejects the genuine correspondence in 2 of 3 configurations, and CKA alone accepts 12. Per-configuration counts are in Appendix F.

X3, the same audit on ESM-2 protein language models. Protein language models are used to read structure and function out of sequence, and correspondence claims between them carry the same risk. We repeat the audit on ESM-2 [Lin et al., 2023] across 3 scale pairs, 320 UniRef50 proteins of 128 residues, 864 interventions. All ESM-2 scales share one amino-acid vocabulary, so outputs are comparable without alignment; the masked model's prediction at a residue is read where the intervention was made; and UniRef50 representatives keep held-out proteins below 50% identity to the fitting set.

X3 also carries positive controls, since a guard that refuses everything scores $\mathrm{NAR} = 0$. An identity map cannot fail unless the code is broken, and an adjacent layer of the same model is a real correspondence that must be found. Both are accepted in every configuration, so $\mathrm{PAR} = 1.00$ over these six evaluations while $\mathrm{NAR} = 0.00$. (X1 has one positive per configuration, the genuine pair.) Guard 1 alone accepts 0.80 of the nulls (12 of 15), so the same failure appears on protein sequences.

A suite of named nulls says only whether those particular candidates got through. We therefore drew 24 fresh members of each of 3 null families (permuted pairings, random rotations and covariance-matched Gaussian targets) for one language and one protein pair. Over 144 draws guard 1 accepts 1.00 and the pair 0.00; the best null reaches a median correlation of 0.09 against the threshold of 0.2 ($\mathrm{AnyNullPass} = 0$).

X2, an SAE feature's meaning. Sparse autoencoders are the standard tool for extracting candidate features from a residual stream [Bricken et al., 2023]. We train a TopK autoencoder ($8\times$, $L_0 = 32$, recovering 93–99% of activation variance depending on the arm) on one model's residual stream. Guard 1 is the correlational reading a top-activating-examples panel supports, the enrichment of a feature's firings on a token type over its base rate. Guard 2 ablates the feature and requires the damage to be specific, degrading prediction on that concept more than elsewhere. The vacuity class is a random direction, a scrambled feature, a correlated surrogate and a near-dead feature.

The descriptive guard behaves differently here. It accepts 0 of 16 null evaluations (four typed controls in each of four arms; $\mathrm{NAR} = 0.00$, AnyNullPass 0). Every null scores enrichment of at most 1.2 against a threshold of 4, so this suite tests little (Appendix F). The guard's failure appears in CAR.

Selection guarantees enrichment. Features are selected as the 150 most enriched per arm (Table 11), so their passing guard 1 is true by construction, and CAR means something only against a matched comparison. For each selected feature we draw three controls from the same arm. They are matched on firing rate to within 20% and on mean activation within a factor of 2, their top concepts differ from the selected feature's, and they are scored on the selected feature's concept. The controls count as supported when at least two of the three are. Of the 600 selected features, 191 have fewer than 30 occurrences of their concept in the consequence split, too few for H (Appendix F), and 4 have no matched control. CAR is over the remaining 405 paired features; over all 409 features with a computable test it is 0.45.

Table 3: X2. Among features selected for strong enrichment an independent intervention supports 0.45; among matched controls scored on the same concepts, 0.005 ($\Delta\mathrm{CAR} = +0.45$, 95% CI $+0.29$ to $+0.58$, positive in every arm). Enrichment therefore carries causal information a matched control does not, and is not well calibrated. The interval is a cluster bootstrap over arms and over features sharing a top token (the 405 features share 22 concept tokens), wider than treating them as independent trials. Rows are generated from the result file.

arm	paired	CAR selected	CAR matched control	ΔCAR
Pythia-160m L4	93	0.24	0.00	+0.24
Pythia-160m L8	111	0.54	0.00	+0.54
Pythia-410m L16	102	0.58	0.00	+0.58
Pythia-410m L8	99	0.42	0.02	+0.40
pooled	**405**	**0.45** (CI 0.29–0.58)	**0.005**	**+0.45**

A feature is supported when $S_f = (\Delta L_C - \Delta L_{\neg C})/\mathrm{SE}[\cdot] > 1.96$ with $\Delta L_C > 0$, where ΔL is the loss increase under ablation on the concept C and elsewhere. The endpoint $\Delta\mathrm{CAR}$, its retraction condition and the statistic S_f were fixed in a timestamped specification before the first matched-control run, which drew one control per feature and gave $\Delta\mathrm{CAR} = +0.39$, positive in every arm. The pipeline was then revised, including the concept definition, the ablation and the controls (Appendix F). With the first of the three controls alone, the control rate is 0.02 and $\Delta\mathrm{CAR} = +0.43$. Table 3 gives the result: the gap to matched controls is positive in every arm and its interval excludes zero, so the guard carries real information. Agreement is far from complete, and with NAR at 0.00 no null control shows this.

The consequence test's own positive control. A low CAR could mean that ablation misses real features. For each selected feature we also ablate the gradient of the concept token's logit with respect to the residual stream. A small change along this direction moves that logit most, and $\mathrm{PAR}(H)$ is the rate at which H supports it. The control is valid where ablating that direction damages prediction of the concept more than prediction elsewhere. In the two Pythia-410m arms it does the reverse for 0.93 and 0.19 of features, so the control is invalid there (Table 12). The positive control and its 0.10 wrong-sign cutoff are not part of the timestamped specification. On the two arms where the control is valid, H detects 0.90 of the gradient directions (184 of the 204 features where it is defined). Validity is judged by the sign of the same score, so this rate is conditional on that selection. On the same arms it supports 0.40 of the 204 paired features, and matched controls 0.000. A feature with a smaller effect than the gradient direction can still be missed.

A second intervention, injecting the feature where it is silent, supports 0.65 of the paired features; counting either intervention as support gives 0.70 (Appendix F).

What this changes. There are two failure modes, and the four physics operations exposed the first (Table 9). The recommendation has two halves: report a guard's behaviour on empty inputs, *and* report how often it agrees with a refutable test, itself checked on positives, on candidates that are not empty.

7 The operations recover what they should

NAR is interpretable only if the operations accept something, and with both guards applied they do. L1 splits five invariant directions on a lattice gauge model into one redundancy and four symmetries, whose charges drift by at most 7.6×10^{-10} against 8.6×10^{-1}. L2 accepts above a separation floor between $12\times$ and $25\times$. L3 accepts real cases $6/6$ and matched controls $0/6$. L4 recovers a hidden carrier of $+1.192$ per event against a withheld 1.200. Appendix B gives the full account.

8 Related work

The principle is not new, and neither are several of the findings. Adebayo et al. [2018] showed several saliency methods produce near-identical maps after the model's weights are randomised. Hewitt and Liang [2019] scored probes by selectivity, accuracy on the real task minus accuracy on a control task with randomised labels. Davari et al. [2023] showed CKA can be moved without changing what a network does, half of our X1; Jacovi and Goldberg [2020] separated faithfulness from plausibility; and SHAPE [Song et al., 2026] describes reasoning traces with an instrument whose own null behaviour is unreported (Appendix C). Vafa et al. [2025] show by probing that a model predicting orbits accurately fails to apply Newtonian mechanics to new tasks, the shape of our L2 finding; Posner et al. [2026] review circuits in single-cell models that track co-expression rather than regulation, the gap X2 measures as a CAR.

Deployed pipelines are built from guards. Foster et al. [2026] choose which candidates are worth executing, where a misallocation is recoverable because whatever runs is checked by its own execution. Zheng et al. [2026] carry 6,453 conjectures down to 77 items for expert review, each stage a guard and the last deciding what a mathematician's attention is spent on. Their prompt instructs the model to check that a repair is not "vacuous, or so weakened that it no longer captures the intended conjecture": our concern, handled by instruction rather than by a measured acceptance rate.

Interventions and benchmarks. Geiger et al. [2021] formalised interchange interventions and Meng et al. [2022] made activation patching standard practice; our second guard belongs to that family. SAEBench [Karvonen et al., 2025] and MIB [Mueller et al., 2025] supply shared metrics, tasks and baselines; the rate at which their criteria accept a candidate built to carry nothing is the quantity a null suite adds.

9 Limitations

The corpus has one annotator. The prize corpus of Appendix G was annotated by one language model under human direction; it motivates the operations and supports no claim here, and we ship a re-annotation kit.

The proposers are narrow. Coarse-graining and transfer search only linear maps, and recall on the proposer's estimate of the true subspace is 0.71 against 1.00 on the subspace itself. L4 posits a quantity of charge, not an entity; L5, a change in what counts as a move, was not attempted; a true reduction tilted by 0.03–0.08 radians is refused; and four of eight concept-forming Standard Model episodes carry no probe for L1 (Appendix E).

The models are small. Pythia models up to 410M parameters on WikiText-2 and ESM-2 up to 150M; larger models are the next measurement.

References

Julius Adebayo, Justin Gilmer, Michael Muelly, Ian Goodfellow, Moritz Hardt, and Been Kim. Sanity checks for saliency maps. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

doi: 10.48550/arXiv.1810.03292.

Anthony N. Aguirre. Dust versus cosmic acceleration. *Astrophysical Journal Letters*, 512(1): L19–L22, February 1999. ISSN 0004637X. doi: 10.1086/311862.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *PMLR*, pages 2397–2430, 2023. doi: 10.48550/arXiv.2304.01373.

Daniil A. Boiko, Robert MacKnight, Ben Kline, and Gabe Gomes. Autonomous chemical research with large language models. *Nature*, 624(7992):570–578, 2023. doi: 10.1038/s41586-023-06792-0.

Trenton Bricken, Adly Templeton, Joshua Batson, et al. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. URL https://transformer-circuits.pub/2023/monosemantic-features.

Seyone Chithrananda, Gabriel Grand, and Bharath Ramsundar. ChemBERTa: Large-scale self-supervised pretraining for molecular property prediction. *arXiv preprint arXiv:2010.09885*, 2020. doi: 10.48550/arXiv.2010.09885.

Miles Cranmer. Interpretable machine learning for science with PySR and SymbolicRegression.jl. *arXiv:2305.01582*, 2023. doi: 10.48550/arXiv.2305.01582.

Andrew Cropper and Sebastijan Dumančić. Inductive logic programming at 30: A new introduction. *Journal of Artificial Intelligence Research*, 74:765–850, June 2022. ISSN 1076-9757. doi: 10.1613/jair.1.13507.

MohammadReza Davari, Stefan Horoi, Amine Natik, Guillaume Lajoie, Guy Wolf, and Eugene Belilovsky. Reliability of CKA as a similarity measure in deep learning. In *International Conference on Learning Representations (ICLR)*, 2023. doi: 10.48550/arXiv.2210.16156.

A. Einstein. Zur Elektrodynamik bewegter Körper. *Annalen der Physik*, 322(10):891–921, January 1905. ISSN 0003-3804, 1521-3889. doi: 10.1002/andp.19053221004.

Kevin Ellis, Catherine Wong, Maxwell Nye, Mathias Sablé-Meyer, Lucas Morales, Luke Hewitt, Luc Cary, Armando Solar-Lezama, and Joshua B. Tenenbaum. DreamCoder: Bootstrapping inductive program synthesis with wake-sleep library learning. In *Proc. 42nd ACM SIGPLAN International Conference on Programming Language Design and Implementation (PLDI)*, pages 835–850, Virtual Canada, June 2021. ACM. ISBN 978-1-4503-8391-2. doi: 10.1145/3453483.3454080.

Scott E. Fahlman and Christian Lebiere. The cascade-correlation learning architecture. In David S. Touretzky, editor, *Advances in Neural Information Processing Systems 2*, pages 524–532, San Mateo, CA, 1990. Morgan Kaufmann. ISBN 978-1-55860-100-0.

Thomas Simon Foster, Bassel Al Omari, Tingchen Fu, Thomas Mann, Carl Domond, Lucia Cipolina-Kun, Bhavul Gauri, Muna Aghamelu, Alexander D. Goldie, Eryk Helenowski, Jean-Christophe Gagnon-Audet, Alberto Pepe, Saba Nazir, Daniel Izcovich, Noam Levi, Rishi Hazra, Karen Hambardzumyan, Nicolas Baldwin, Xian Li, Martin Josifoski, Paris Giampouras, Masoud Jalili Sabet, Anya Sims, Hela Momand, Tatiana Shavrina, Despoina Magka, Jason Weston, Yulin Wang, Anirudh Goyal, João Henriques, Yoram Bachrach, Emily McMilin, and Jakob Nicolaus Foerster. AI research preference models. *arXiv preprint arXiv:2608.13940*, 2026. doi: 10.48550/arXiv.2608.13940.

Atticus Geiger, Hanson Lu, Thomas Icard, and Christopher Potts. Causal abstractions of neural networks. In *Advances in Neural Information Processing Systems 34 (NeurIPS 2021)*, 2021. doi: 10.48550/arXiv.2106.02997.

William Gilpin. Chaos as an interpretable benchmark for forecasting and data-driven modelling. In *Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track*, 2021. doi: 10.48550/arXiv.2110.05266.

Arya Grayeli, Atharva Sehgal, Omar Costilla-Reyes, Miles Cranmer, and Swarat Chaudhuri. Symbolic regression with a learned concept library. In *Advances in Neural Information Processing Systems 37*, pages 44678–44709, Vancouver, BC, Canada, 2024. Neural Information Processing Systems Foundation, Inc. (NeurIPS). ISBN 979-8-3313-1438-5. doi: 10.52202/079017-1419.

John Hewitt and Percy Liang. Designing and interpreting probes with control tasks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2733–2743, 2019. doi: 10.18653/v1/D19-1275.

Alon Jacovi and Yoav Goldberg. Towards faithfully interpretable NLP systems: How should we define and evaluate faithfulness? In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 4198–4205, 2020. doi: 10.18653/v1/2020.acl-main.386.

Lou Jost. Entropy and diversity. *Oikos*, 113(2):363–375, 2006. doi: 10.1111/j.2006.0030-1299.14714.x.

John Jumper, Richard Evans, Alexander Pritzel, Tim Green, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596(7873):583–589, 2021. doi: 10.1038/s41586-021-03819-2.

Adam Karvonen et al. SAEBench: A comprehensive benchmark for sparse autoencoders in language model interpretability. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, 2025. doi: 10.48550/arXiv.2503.09532.

Ross D. King, Kenneth E. Whelan, Ffion M. Jones, Philip G. K. Reiser, Christopher H. Bryant, Stephen H. Muggleton, Douglas B. Kell, and Stephen G. Oliver. Functional genomic hypothesis generation and experimentation by a robot scientist. *Nature*, 427(6971):247–252, January 2004. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature02236.

Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, volume 97 of *PMLR*, pages 3519–3529, 2019. doi: 10.48550/arXiv.1905.00414.

Pat Langley. Data-driven discovery of physical laws. *Cognitive Science*, 5(1):31–54, 1981. doi: 10.1111/j.1551-6708.1981.tb00869.x.

Federico Lelli, Stacy S. McGaugh, and James M. Schombert. SPARC: Mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves. *Astron. J.*, 152(6):157, 2016. doi: 10.3847/0004-6256/152/6/157.

Zeming Lin, Halil Akin, Roshan Rao, Brian Hie, Zhongkai Zhu, Wenting Lu, Nikita Smetanin, Robert Verkuil, Ori Kabeli, Yaniv Shmueli, Allan dos Santos Costa, Maryam Fazel-Zarandi, Tom Sercu, Salvatore Candido, and Alexander Rives. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

H. A. Lorentz. Electromagnetic phenomena in a system moving with any velocity smaller than that of light. *Proc. R. Neth. Acad. Arts Sci.*, 6:809–831, 1904. URL https://www.dwc.knaw.nl/DL/publications/PU00014148.pdf. Reprinted in *Collected Papers*, Vol. 5, 1937.

Chris Lu, Cong Lu, Robert Tjarko Lange, Yutaro Yamada, Shengran Hu, Jakob Foerster, David Ha, and Jeff Clune. Towards end-to-end automation of AI research. *Nature*, 651:914–919, 2026. doi: 10.1038/s41586-026-10265-5.

Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. doi: 10.52202/068431-1262.

M. Milgrom. A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis. *Astrophys. J.*, 270:365–370, 1983. doi: 10.1086/161130.

Iman Mirzadeh, Keivan Alizadeh, Hooman Shahrokhi, Oncel Tuzel, Samy Bengio, and Mehrdad Farajtabar. GSM-Symbolic: Understanding the limitations of mathematical reasoning in large language models. In *International Conference on Learning Representations (ICLR)*, 2025. doi: 10.48550/arXiv.2410.05229.

Aaron Mueller, Atticus Geiger, Sarah Wiegreffe, Dana Arad, Iván Arcuschin, Adam Belfki, Yik Siu Chan, Jaden Fiotto-Kaufman, Tal Haklay, Michael Hanna, Jing Huang, Rohan Gupta, Yaniv Nikankin, Hadas Orgad, Nikhil Prakash, Anja Reusch, Aruna Sankaranarayanan, Shun Shao, Alessandro Stolfo, Martin Tutek, Amir Zur, David Bau, and Yonatan Belinkov. MIB: A mechanistic interpretability benchmark. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 45069–45108, 2025. doi: 10.48550/arXiv.2504.13151.

OpenAI. gpt-oss-120b and gpt-oss-20b model card, 2025.

Linus Pauling. Apparent icosahedral symmetry is due to directed multiple twinning of cubic crystals. *Nature*, 317(6037):512–514, October 1985. ISSN 0028-0836, 1476-4687. doi: 10.1038/317512a0.

S. Perlmutter et al. Measurements of Ω and Λ from 42 high-redshift supernovae. *Astrophysical Journal*, 517(2):565–586, June 1999. ISSN 0004-637X, 1538-4357. doi: 10.1086/307221.

Ingmar Posner, Anson Lei, and Bernhard Schölkopf. From observation to insight: Mechanistic world models and the quest for autonomous discovery. *arXiv preprint arXiv:2607.12474*, 2026. doi: 10.48550/arXiv.2607.12474.

Maithra Raghu, Thomas Unterthiner, Simon Kornblith, Chiyuan Zhang, and Alexey Dosovitskiy. Do vision transformers see like convolutional neural networks? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. doi: 10.48550/arXiv.2108.08810.

A. Raue, C. Kreutz, T. Maiwald, J. Bachmann, M. Schilling, U. Klingmüller, and J. Timmer. Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, August 2009. ISSN 1367-4811, 1367-4803. doi: 10.1093/bioinformatics/btp358.

Adam G. Riess et al. Observational evidence from supernovae for an accelerating universe and a cosmological constant. *Astronomical Journal*, 116(3):1009–1038, September 1998. ISSN 00046256. doi: 10.1086/300499.

Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, Matej Balog, M. Pawan Kumar, Emilien Dupont, Francisco J. R. Ruiz, Jordan S. Ellenberg, Pengming Wang, Omar Fawzi, et al. Mathematical discoveries from program search with large language models. *Nature*, 625(7995):468–475, 2024. doi: 10.1038/s41586-023-06924-6.

Michael Schmidt and Hod Lipson. Distilling free-form natural laws from experimental data. *Science*, 324(5923):81–85, 2009. doi: 10.1126/science.1165893.

D. Shechtman, I. Blech, D. Gratias, and J. W. Cahn. Metallic phase with long-range orientational order and no translational symmetry. *Physical Review Letters*, 53(20):1951–1953, November 1984. ISSN 0031-9007. doi: 10.1103/PhysRevLett.53.1951.

Jonghyun Song, Sangjun Song, Minjae Oh, Haesung Pyun, Sungsik Lee, and Yohan Jo. SHAPE of chain-of-thought in math reasoning. In *The 3rd AI for Math Workshop at the 43rd International Conference on Machine Learning (ICML)*, 2026. doi: 10.48550/arXiv.2608.28600.

Trieu H. Trinh, Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. Solving olympiad geometry without human demonstrations. *Nature*, 625:476–482, 2024. doi: 10.1038/s41586-023-06747-5.

Silviu-Marian Udrescu and Max Tegmark. AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16):eaay2631, April 2020. ISSN 2375-2548. doi: 10.1126/sciadv.aay2631.

Keyon Vafa, Peter G. Chang, Ashesh Rambachan, and Sendhil Mullainathan. What has a foundation model found? Using inductive bias to probe for world models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 60727–60747, 2025.

Zeyu Zheng, Shengtong Zhang, Jeremy Avigad, Prasad Tetali, and Sean Welleck. The problem is the problem: Towards scalable mathematical discovery. *arXiv preprint arXiv:2608.16977*, 2026. doi: 10.48550/arXiv.2608.16977.

Robert Zwanzig. Ensemble method in the theory of irreversibility. *J. Chem. Phys.*, 33(5):1338–1341, 1960. doi: 10.1063/1.1731409.

A The four operations in full

L1, reinterpretation. For a theory with parameters p predicting $f(x, p)$, the directions along which no prediction moves are the null space of $J_{ij} = \partial f_i / \partial p_j$: structural non-identifiability [Raue et al., 2009], read as a property of a theory rather than a nuisance in a fit. The obvious move is to quotient that null space away. *This is wrong*: a global phase rotation is invisible to every probe exactly as a gauge transformation is, and removing both erases the conserved quantity instead of discovering it. Three tests must be kept apart — **probe invariance**, **dynamical invariance**, and the **local versus uniform** distinction that separates gauge redundancy from a global symmetry (Table 4). We do not claim probe invariance yields a Noether charge in general; within this constructed model the local test is what classifies one.

L1 on data with error bars. Exact degeneracy is the wrong test for real data: Lorentz's aether was not unobservable in principle, only to within the precision of the day. We scale column j of the Jacobian by $|p_j|$ and row i by $1/\sigma_i$, and ask whether a direction moves by an order-unity amount without shifting any prediction past its error bar. The second guard is the locality test of Table 4.

Table 4: The locality test: one null-space computation, two different moves. This is the second guard for L1, and the difference between deleting a structure and discovering a conservation law.

invariant *point by point*	redundancy: quotient it	[L1]
invariant only *uniformly*	symmetry: keep it, read off a conserved charge	[L4]

A direction that no probe sees when it is moved uniformly, but that probes do see when it is moved locally, is a global symmetry and carries a conserved charge. The implementation refuses to quotient at all when no locality test is supplied: `quotient()` raises rather than proceeding, because the failure mode it guards against is silent.

L2, coarse-graining. Closure is scored by integrating the coarse law alone and comparing with the truth, normalised by the trajectory's own spread. The *derived* law projects the known fine law through the reduction. The *learned* law is a polynomial regression from coarse states to coarse velocities, with velocity targets taken from the fine law at the lifted state. It has linear and cubic terms on the lattice and terms up to quadratic on the benchmark. Which law carries the guard depends on the experiment. On the slow–fast systems (Table 5, the separation sweep and the labelled suite of Sec. 5) closure uses the derived law, which has nothing fitted. On the lattice both laws are computed and the better one is used. The learned law there is fitted on every trajectory, including the scored one. Both choices favour acceptance, and the lattice reduction is refused under both. On the external benchmark acceptance rests on the learned law alone, fitted on trajectories other than the scored one. With both laws reported, a negative can be traced to the reduction or to the fitter. The pay guard is fit-free by construction, after three fitted versions failed in three different ways, none of which produced an obviously wrong number. Its hard invariant is that a reduction keeping everything must score exactly 1.000.

Table 5: Three vacuity controls on the slow–fast system at $\epsilon = 0.02$. Two of the three close *perfectly*; only the pay guard of Eq. 4 refuses them, far below its 0.50 threshold. Pay values are from the covariance-weighted score; the unweighted version this replaced drove both empty cases to the -1 clip, which reported saturation rather than a measurement.

control	closes	pays	refused by
constant reduction (every state to zero)	**0.000**	-0.54	guard 2 only
random subspace	2.167	0.18	guard 1
the **fast** modes instead of the slow ones	**0.000**	-0.50	guard 2 only

Properties of the pay score. Write $R = Q^{\dagger}Q$ for the reconstruction and $\mathcal{E}$ for the state ensemble. Three properties follow directly from Eq. 4, and the first and third are asserted in the shipped code.

1. **Identity.** If $R = I$ the numerator vanishes and $\mathrm{Pay} = 1$ exactly. This is the invariant that caught three earlier fitted versions of the guard, none of which reached it.
2. **Total loss.** If R maps every state to one point, $P(R\phi)$ is constant, the numerator becomes $\sum\|P(\phi) - P_0\|^2$ for a fixed P_0, which is at least the denominator, so the ratio is ≥ 1 and the score is at most 0 — reaching the -1 clip when P_0 is far from $\bar{P}$. Negative values therefore mean *worse than predicting the ensemble mean*, and the clip exists so that distance is not mistaken for a measurement.
3. **Any invertible linear map of the probes.** For $P \mapsto AP$ with A invertible, the covariance becomes $A\Sigma A^{\top}$, so each Mahalanobis term $v^{\top}\Sigma^{-1}v$ is unchanged and Pay is unchanged with it. This covers scalar rescaling, unequal per-component rescaling $P \mapsto \Lambda P$, and any change of basis. Translation $P \mapsto P + b$ is likewise absorbed, since the numerator is a difference and the denominator is measured about the mean. The score is therefore a property of the reduction and the probe *span*, not of the units or coordinates the probes are written in.

   The unweighted version this replaced had none of that: it was invariant under a scalar c only, so an observable with a larger numerical scale dominated the sum, and a threshold could not be carried between systems whose probes had different units. Destroying a large-scale component then scored 0.00 while destroying a small-scale one scored 1.00 for the same amount of damage; weighted they score 0.66 and 0.67. `verify.py` asserts both this and $\mathrm{Pay}(I) = 1$.

Why the pay guard is not optional under a parsimony objective. The constant reduction is the most parsimonious candidate there is: one constant variable and the mechanism $\dot{q} = 0$. An objective that scores a coarse description by its simplicity and by how well it predicts its own future therefore ranks it above every real reduction, since it is the global minimiser of both terms. The mechanistic world-model objective of Posner et al. [2026], their Eq. 9, combines a complexity penalty on variables, mechanisms and types, a reuse term on the binding structures across environments, and a prediction loss whose target Eq. 9 leaves unstated; their Sec. 2 describes prediction as inferring future observations. The objective has exactly that form when the prediction loss is taken in the learned coordinates rather than on decoded observations, and the reuse term does not change the ranking, since one mechanism serving every environment is maximal reuse. They ask that variables minimise complexity while maximising explanatory scope. Closure scores neither half; the pay guard of Eq. 4 scores the second, and Sec. 5 measures what closure accepts on 129 systems without it.

Charge loss is reported, never blocked. Discarding state detail can destroy a conserved quantity while still closing. Physics does this deliberately, so the mechanism states which charges a reduction loses rather than refusing it; on the lattice reduction both flavour charges are lost (drift 7.5×10^{-1} and 1.9×10^{0}), and it says so.

L3, transfer. Transfer finds a map carrying one domain's states into another's, so that what the first knows becomes a claim about the second. The structural guard requires a fitted linear map to make the target's probes agree with the source's to within 5%, and is scored on states the map was *not* fitted to. The second guard pushes the source's conserved charge through that map and requires the target's own evolution to conserve it to 10^{-3}. Table 6 gives six seeds, each with a matched control whose symmetry is deliberately broken.

L4, positing an entity. L4 consumes what L1's symmetry branch produces: when a found charge fails to balance across an observed process, the missing amount describes something not yet named. Writing ΔQ for charge in minus charge out, the operation posits a carrier of ΔQ when $\Delta Q \neq 0$ and is silent otherwise. The second guard requires $|\Delta Q|$ to exceed the *noise scale*, not merely to sit many standard errors from zero. It fires only when the missing charge exceeds three times the 2% measurement noise, that is 6% of the input charge. Its nulls have nothing hidden and the measured output miscalibrated by 2.5% and 5%.

Table 6: L3 transfer across six seeds, each with a matched control whose symmetry is deliberately broken. The structure test is scored both on the 40 states the map was fitted to and on 40 disjoint states; only the held-out column is used to accept. Scoring it in-sample, as an earlier version of this work did, overstates its failure rate.

	structure, in-sample	structure, **held out** (< 0.05)	charge test ($< 10^{-3}$)
real	median 0.0057 — 6/6	median 0.0061, worst 0.0463 — **6/6 pass**	median 1.3×10^{-11}, worst 5.9×10^{-11} — **6/6 pass**
control	median 0.0455 — 5/6	median 0.0520, best 0.0405 — **2/6 pass**	median 5.2, best 2.8 — **0/6 pass**

B Results and controls in full

The four operations have so far appeared only as things that fail in a particular way. They also work, and NAR is only interpretable if they do — a guard pair that accepted nothing would score $\mathrm{NAR} = 0$ while being useless, which is the reading Table 14 was built to exclude. This section is what the operations recover when both guards are applied. Negative results are reported as found, and twenty-one engineering checks on cases whose answers we supplied all pass.

L1: recovering and classifying a hidden symmetry from a supplied generator basis. On a periodic 6×6 lattice carrying a complex two-flavour matter field and U(1) links, probed only by gauge-invariant, flavour-blind quantities, the mechanism is handed eight candidate generators and told nothing about which are real. It **finds five invariant directions** and, by the locality test, splits them into **one redundancy** and **four symmetries**, emitting four conserved charges with drift 1.8×10^{-11} to 7.6×10^{-10} against the redundancy's would-be charge at 8.6×10^{-1} — nine orders of magnitude, at the trajectory's own precision. At uniform order the gauge transformation and the global phase act identically; only the local test separates them, and without it all four charges are destroyed to remove one redundancy.

L2, L3, L4. The reduction is accepted at $62.9\times$ and $25.2\times$ separation and refused below, placing the floor between $12\times$ and $25\times$. On the lattice, keeping 6 of 72 modes, it accepts *nothing*, with learned and derived closure agreeing to within 0.002 (1.628 against 1.626), so that negative is the physics rather than our fitting. Transfer accepts real cases $6/6$ and matched controls $0/6$. Positing recovers a hidden carrier of $+1.192$ per event against a withheld truth of 1.200. It is silent when the books balance, when the imbalance is noise, and when the measured output is miscalibrated by a few percent.

Appendix B.1 reports what happens when L1 is applied to a live astrophysical controversy with no known answer: the criterion is *silent*, because it needs two descriptions that agree on the data and these disagree by ordinary model comparison.

Engineering checks (21/21). Every case has an answer we supplied, so passing shows the code does what it claims. The mechanism finds the unobservable direction in a description whose concepts are never separable; quotients it away with predictions unchanged to 10^{-6}; **declines to act** on an honest theory with nothing hidden; **rejects** a patch whose new concept moves no prediction, which is the property that stops "add a parameter" from being free; and accepts a new concept only when the residual really has structure the old language lacked.

L1 in full. The uniform Jacobian has rank 3 of 8, so five directions are invariant; a link phase, a rescale and a single-flavour shift are rejected. The local test then isolates one redundancy (local response 2.6×10^{-5}) from four symmetries (5.1–6.6×10^{1}).

L2 and L4 sensitivity. A mechanism that accepted at every separation would be broken; this one has a measurable floor between $12\times$ and $25\times$. For L4, a real unseen product carrying 2.5% or 5% of the charge gives an imbalance of 16σ or 31σ, and the operation stays silent because both sit below its 6% threshold. This is the cost of the threshold: a carrier that small is missed. The energy deficit in beta decay, the case the operation is modelled on, was far larger than a few percent.

The proposer is the limiting component. Across three runs, a world whose residual hides $0.4 \sin 3x$ yielded eight proposals of which two to five were accepted; a control whose residual is pure noise yielded eight proposals of which **none** was accepted, in every run. Offered the true concept directly the filter separates cleanly: $\sin 3x$ reaches 97% coverage against 68% for $\sin 2.9x$ and 20% for wrong terms. In the world hiding $\sin 3x$ the model offered $\sin \pi x$, $\sin 2\pi x$, $\sin 3.7x$ and related terms, never $\sin 3x$.

B.1 Where the criterion goes silent, on real data

We applied L1 to a live controversy with no known answer: the rotation curve of NGC 3198 from SPARC [Lelli et al., 2016], fitted by a dark-matter halo and by MOND [Milgrom, 1983] with its acceleration scale fixed and free. All three descriptions carry *zero* unobservable directions — the halo model's softest direction still moves the curve 6.4σ under a 100% excursion, unsurprising since SPARC's photometry was assembled to break exactly that degeneracy. **The criterion is silent.** Its signature needs two descriptions that *agree* on the data; these disagree at χ^2/N of 1.22 against 9.21, where ordinary model comparison settles it. The operation applies to a narrower class of disputes than we assumed.

C The audit protocol, stated

Eq. 1 and Eq. 2 define what we report. Four things need saying about them.

These are estimates against a list, not worst cases. An earlier version of this paper defined the quantity as a supremum, $\sup_{c \in \mathcal{V}} \Pr[G(c; D) = 1]$, while reporting fractions of controls accepted. Those disagree: for a deterministic guard the supremum is 1 as soon as any null passes. NAR is the mean we were computing all along and AnyNullPass is what the supremum was reaching for, so both are now reported and neither is called a worst case. We do not optimise over a candidate family, so no supremum claim would be warranted.

V must be enumerated and typed — and we chose it. A guard with $\mathrm{NAR} = 0$ against a weak V has been told nothing, so we list the suite and label each member by what it preserves and destroys (Table 7).

Table 7: The null suite is typed rather than pooled: a guard that survives a random direction but not a structure-preserving one has told us something specific, and pooling them into one rate hides it.

type	preserves	destroys
DEGENERATE	nothing	all structure (constant reduction, zero map)
RANDOMISED	dimensionality	any relation to the source
CORRESPONDENCE-BREAKING	marginal statistics	the pairing between source and target
STRUCTURE-PRESERVING	the invariant itself	only its redundancy status
TASK-CONFOUNDED	a trained model's geometry	the semantic content of the input
SUB-THRESHOLD	an effect's significance	its size against the noise scale

Choices made after a first run. X1's first three nulls gave guard 1 an NAR of 0.00. We judged them too weak and added stronger ones. The current suite (Table 2) has since replaced all of them with correspondence-breaking nulls that share the vocabulary (Appendix F). Stronger nulls raise guard 1's NAR and make guard 2's test harder. X2's selection rule changed after a first attempt returned features whose top tokens occurred 3 to 29 times, and its analysis changed after its first matched-control run (Appendix F). Per-control results are reported throughout, so a reader can recompute NAR over any sub-suite.

How diverse is a null suite? Counting nulls overstates a suite whose mass sits on one construction: six nulls that are five variants of one degenerate construction are not six independent ways of being empty. We therefore report the effective number of types a suite exercises, $\exp$ of the entropy of its type mix [Jost, 2006], which falls toward 1 as one type dominates and reaches the nominal count only when types are balanced. Our suites give 3.79 of 4 types for X1, 2.83 of 3 for X2, and 2.59 of 3 for X3, so no headline NAR here is carried by a single construction. Song et al. [2026] use the same

statistic for the reverse purpose, to measure how an LLM spreads reasoning effort over interpretations of a problem.

The vacuous repair, as a null type. Our null types are degenerate, randomised, correspondence-breaking, structure-preserving, task-confounded and sub-threshold. Zheng et al. [2026] name a specific member of the structure-preserving family that we had not, and that generalises well beyond their setting: the *vacuous repair*. When a candidate fails, a discovery system may weaken it until it is trivially true rather than abandon it, and the weakened statement keeps the objects, the notation and the shape of the original while carrying none of its content. Their pipeline guards against this by instructing the prover to check that a repair is not "so weakened that it no longer captures the intended conjecture".

This is the same failure as the constant reduction of Sec. 4, in a different domain. A reduction mapping every state to zero is a coarse-graining weakened until closure is trivial; a vacuously repaired conjecture is a statement weakened until proof is trivial. Both keep the form and lose the content, and both are accepted by a guard that scores form. The recipe transfers: for any operation that can weaken a candidate rather than reject it, the weakened-to-triviality version belongs in V, and its acceptance rate is the quantity to report. We recommend it as a default member of the null suite for systems that repair candidates, and note that we have not run it here, since none of our four operations repairs a failed candidate.

A descriptive instrument that states its own gap. SHAPE labels chain-of-thought traces with mathematical heuristics and tracks the *semantic space* the model is working in, then reports how reasoning effort is distributed over those spaces. Its heuristic tagger is validated against an adjudicated reference set. Its semantic-space tracker is not, and the authors say why: the spaces are trajectory-relative interpretations for which, in their words, no objective gold standard exists, so the tracker is calibrated by manual review instead. By the distinction drawn below (the two kinds of guard) it is a structural score, computed on the trajectory it is describing and committing to nothing that could fail, and the paper's structural findings rest on it. What would test it is a null suite: trajectories carrying no coherent interpretation, on which a tracker that labels whatever it is given would still return a well-formed sequence of spaces. **We ran that audit and it was inconclusive**, and we report what running it required. Using the authors' own coding manuals and metric code, on 24 MATH-500 problems, we compared a model's own trace against three typed nulls (its content units reordered, a trace for a different problem, units spliced from unrelated traces) and one positive control (two independent solutions to the same problem concatenated, which contains a genuine restart). Our annotator is a 7B model, not the 27B one they use, because that model does not fit the hardware here. At that scale the tracker takes 19 NEW decisions and 0 RETURN decisions across 120 trajectories. Every condition sits between 1.0 and 1.3 effective spaces, and nothing separates: the positive control itself reaches AUROC 0.39. That is a floor effect in our annotator, not a finding about their tracker, and the null AUROCs of 0.52 to 0.59 carry no information. The audit is shipped so that anyone with the 27B annotator can run it in one command.

The two kinds of guard. A **structural score** $S(c, D)$ is computed on the data the candidate was fitted to; a **consequence test** $H_c(D_{\text{test}})$ requires the candidate to commit in advance to an outcome on information not used to fit it. The evidence here supports a conditional claim rather than a general one, and the condition is not the one we first proposed. We expected S-guards to have high NAR only *when the candidate has freedom relative to* D, so that more data would fix them. That holds for the fitted map: scored in-sample, R^2 accepts 0.80 of X1's nulls at eight stimuli and 0.20 at 8,192. It does not explain the held-out figure of 0.60, or 0.80 for the protein models. The residue is a second mechanism entirely: linear CKA is a function of the two sets of representations, so it is invariant to *every* claim about how they correspond, and no amount of data touches that. X2's co-occurrence count has little freedom and no such invariance, and reaches 0.00 against a weak suite (Appendix F). Consequence tests fail in other ways, and reporting both kinds of guard separates the two.

Low NAR is necessary, not sufficient. X2 is the counterexample: a guard rejecting every null still leaves 55% of what it accepts unsupported by the ablation test (Table 3), and 0.30 unsupported by either of two interventions. A complete report of a guard needs NAR against a typed suite *and* CAR against a consequence test fixed before it is scored.

Table 8: The invariants that caught our own wrong guards. A version violating one of these is wrong however plausible its numbers look; the first two found three earlier pay guards and an early causal guard, and the third found L3's in-sample structural score.

a reduction that keeps everything	must score exactly 1.000
a guard	must never rank a null candidate above the real one
a structural score	must be reported on data the candidate was not fitted to

Table 9: The two failure modes of Sec. 2. The physics operations and X1 and X3 show the first; X2 shows the second.

	the guard accepts…	seen in
null acceptance (vacuity)	candidates carrying nothing the operation seeks	L1–L4, X1, X3
unsupported acceptance	candidates an independent test does not support	X2 (405 paired features)

D A language model on the reinterpretation task

A language model can recount the Lorentz–Einstein episode, so it is fair to ask whether the operation is already available in a system that has read the histories. We put five episodes, three historical and two constructed controls, to a 20B open-weight model [OpenAI, 2025] in three conditions: **named**; **blind**, every identifier alpha-renamed with fit statistics retained; and **bare**, renamed with statistics withheld. Table 10 breaks the result out by episode.

Table 10: Accuracy at identifying which of two candidates is the reinterpretation. Twelve repeats per cell, presentation order randomised, chance 50%.

episode	named	blind	bare
dark energy [Riess et al., 1998, Perlmutter et al., 1999] vs. grey dust [Aguirre, 1999]	67%	42%	50%
relativistic mass vs. Lorentz's aether [Lorentz, 1904, Einstein, 1905]	100%	100%	100%
quasicrystals [Shechtman et al., 1984] vs. twinning [Pauling, 1985]	100%	50%	42%
two constructed controls	83%	25%	38%
pooled, 60 trials	**87%**	48%	53%

Accuracy is far above chance when the model can recognise the episode ($p = 5{\times}10^{-9}$) and *indistinguishable from chance* when it cannot ($p = 0.9$ blind, 0.7 bare; named against blind, paired $p = 0.03$). On these five episodes it behaves as though recalling rather than judging. We do not claim this generalises beyond them, and renaming alone is already known to move accuracy on grade-school arithmetic [Mirzadeh et al., 2025]. One case survives anonymisation: relativity stays at 100% because $(1 - v^2)^{-1/2}$ is recognisable from its algebra whatever the identifiers are called. Withholding or supplying the statistics rescues neither, so this is not a missing-information problem.

E What probe-based discovery cannot reach

The machinery above buys concepts with measurements. Tested against eight concept-forming episodes in the Standard Model it accounts for **two**: the neutrino, forced by a continuous spectrum where a two-body decay demands a line, and gauge redundancy. **Four of the other six contain no probe at all.** Colour was invented so the Δ^{++} state would be *allowed* under Fermi statistics; charm so that flavour-changing neutral currents would *cancel*; complete generations so that gauge anomalies would sum to zero; and the Higgs below about a TeV so that WW scattering would stay unitary — which told experimenters what to build.

Each predicted a real particle, and each was justified by a **theorem first**, with the experiment arriving years or decades later. A framework that pays for a concept with an experiment does not reach these. Reaching them needs a formal theory in which to prove theorems, and no discovery system we know of carries one. We see this as a main limit on automated physics discovery today.

F The three interpretability experiments in full

X1 and X2 use models from the Pythia suite [Biderman et al., 2023], whose members share a tokenizer, which is what makes the causal comparison in X1 possible at all; their text is WikiText-2. X3 uses ESM-2. Everything is scored on held-out data.

X1 setup. 4 configurations over the Pythia suite: 70m L4→160m L8, 70m L2→160m L4, 160m L8→410m L12 and 70m L4→410m L12. Text is 240 WikiText-2 sequences of 128 tokens; the map is fitted on half and everything is scored on the other half. Guard 1 is ridge regression ($\lambda = 10^2$) reporting held-out R^2 about the *fitting* mean, plus linear CKA, accepting at $R^2 > 0.5$ or $\mathrm{CKA} > 0.5$. Guard 2 runs 480 interventions per configuration (5 seeds $\times$ 8 positions $\times$ 12 donors, each scored over 16 recipient sequences). Each adds the mapped difference at a single position and correlates the two models' induced shifts after centring across donors. Guard 2 accepts when the median correlation exceeds 0.2, and a candidate is accepted when both guards accept.

Each candidate is scored with the map it claims. A candidate that composes the fitted map with a rotation is a claim about *that* composed map, and an earlier version scored every candidate with the real map, so the rotated and donor-permuted rows reported R^2 identical to the genuine pair. Scored correctly the rotation falls to -0.96, while its CKA does not change.

An earlier pathology, and what removed it. We previously reported that our own causal guard preferred a vacuous candidate: correlating *uncentred* shifts ranked a random-token control at 0.70 against 0.39 for the genuine pair, and centring across donors was what fixed it. That was an artefact of the intervention rather than of the metric. The old guard overwrote the residual stream at every position with one mean-pooled donor vector, which pushed both models toward the same frequent tokens, and that shared generic response dominated the correlation. Under localised difference patching the generic response is small: the worst null now scores 0.23 uncentred against 0.67 for the weakest genuine pair. Centring is still applied, and is no longer load-bearing. We report this because the earlier claim appeared in a previous version of this work.

The earlier null suite also included ChemBERTa [Chithrananda et al., 2020] reading SMILES as a different-domain control, for which guard 2 was *undefined*: the causal test compares two next-token distributions and a chemistry model shares no vocabulary with an English one. The rebuilt suite replaces it with correspondence-breaking nulls that share the vocabulary, which is a harder test, and the limitation it exposed stands: the refutable guard needs a shared output space to compare in.

X3 setup. 3 ESM-2 [Lin et al., 2023] scale pairs (8M L3→35M L6, 8M L4→150M L15 and 35M L6→150M L15) on 320 UniRef50 sequences truncated to 128 residues, half fitted and half held out, 288 interventions per configuration. Guards are *imported* from the X1 module, not reimplemented. Two candidates are positive controls rather than nulls: an identity map at the same layer, which cannot fail unless the code is broken and is there to check exactly that, and the adjacent layer of the same model, which is a genuine correspondence the guard must find. Both are accepted in every configuration.

X2 setup. One TopK sparse autoencoder per arm, on Pythia-160m layers 4 and 8 and Pythia-410m-deduped layers 8 and 16, 8$\times$ expansion (6,144 or 8,192 features), $L_0 = 32$, recovering 93–99% of activation variance depending on the arm; 1,200 sequences split 60/20/20 into autoencoder training, guard scoring and the consequence test, so a feature is selected on one split and judged on another. An L_1 penalty was tried first and would not sparsify — at the coefficient that left reconstruction usable, L_0 settled near 565 and the most frequently firing features were generic ones whose top token was simply the corpus's most common token. TopK sets L_0 by construction.

Two measurement pitfalls we hit, both of which inflate guard 1. First, ranking features by raw firing frequency selects the always-on ones; every "most interpretable" feature it returned had *the* as its top token. Second, and less obvious: scoring a feature by the likelihood ratio $P(\text{token} \mid \text{fires})/P(\text{token})$ is degenerate when the feature captures every occurrence of a rare token, because the ratio then collapses to $N/(\text{times fired})$ and is independent of how many occurrences there were. The twelve "most selective" features it first returned had top tokens occurring 3 to 29 times in held-out

text — too few to intervene on. We therefore require a candidate concept to occur at least 60 times in the guard split before it is scored.

X2 nulls. In each arm guard 1 scores four nulls. The first is a random unit direction in the residual space. The second is the decoder vector of the arm's most enriched feature with its coordinates permuted, and the third is the same vector, unpermuted, plus Gaussian noise of 0.6 times its norm. The fourth is the feature with the lowest nonzero firing rate. The first three are read as the residual stream's projection on the direction, clipped at zero. All sixteen score enrichment of at most 1.2 against the threshold of 4. The surrogate keeps little of its source feature's enrichment (1.0–1.2 against 137–187), so this suite tests little. In the first run to the specification, under the earlier concept definition below, the surrogate passed guard 1 in one arm (enrichment 69, $\mathrm{NAR} = 0.06$).

Full X2 result. Table 11 gives the selection flow. A feature is selected at enrichment > 4; support requires the standardised difference $S_f = (\Delta L_C - \Delta L_{\neg C})/\mathrm{SE}[\cdot]$ to exceed 1.96 with $\Delta L_C > 0$. An earlier version thresholded the raw ratio $\Delta L_C/\Delta L_{\neg C}$, which is unstable when the denominator is near zero, where one feature scored 1644. The standardised difference replaces it. This is stricter, and the reported CAR is correspondingly lower than the ratio gave.

Table 11: X2 selection flow. Features are ranked by enrichment and the top 150 per arm are taken, so the fact that every selected feature clears the threshold is a property of the pipeline and is not reported as a result. Between 605 and 1453 features per arm clear it; we test 150. Rows are generated from the result file.

stage	160m L4	160m L8	410m L16	410m L8
all SAE features	6144	6144	8192	8192
firing rate in range	3772	1248	2515	3540
concept frequent enough	3772	1248	2508	3540
enrichment > 4	1263	605	1169	1453
selected for intervention	150	150	150	150

The frozen specification and what changed after it. `corpus/ANALYSIS.md` was committed on 1 September 2026, before the first matched-control run. It fixes the endpoint $\Delta\mathrm{CAR}$ and its retraction condition: if the interval includes zero, the X2 claim is withdrawn. It also fixes the support statistic S_f, the selection rule and the matching rule. That run drew one control per feature, matched on firing rate, and gave CAR 0.46 against 0.07 over 600 features: $\Delta\mathrm{CAR} = +0.39$, positive in 4 of 4 arms. The pipeline was then revised, each change made after that result was known. (i) The data were split three ways, so that a feature's concept is chosen on one split and judged on another. (ii) A feature's concept became the next token; it had been the token at the firing position. This changes both guard 1's concept and H's mask. (iii) The ablation became the subtraction of the feature's own contribution; it had erased the whole decoder direction. (iv) The standard error in S_f became one over per-sequence contrasts; it had been over tokens. (v) Each feature received three controls, also matched on mean activation, with the majority rule of Sec. 6. A control counts as supported when $S_f > 1.96$. The first control alone gives a control rate of 0.02 and $\Delta\mathrm{CAR} = +0.43$ (95% CI $+0.27$ to $+0.57$). (vi) The interval resamples arms and concept clusters with each verdict held fixed; the specification also resampled sequences, which counts the noise in S_f twice. (vii) Features too rare in the consequence split to be judged were removed from the denominator, as the specification's exclusion rule requires. Counting them as unsupported gives CAR 0.31. The specification states support as a bootstrap interval that excludes zero; both runs use $S_f > 1.96$. The positive control on H is not in the specification. The first run's result file ships as `xai_sae_spec_run.json`.

The three splits are contiguous blocks of WikiText-2, so they hold different articles. Of the 191 untestable features, 141 have no occurrence of their concept in the consequence split and 50 have at most 24 positions, against the 30 required; the smallest testable concept has 31 positions. 118 of the 191 share one concept token, the "send" of "Townsend", which occurs 140 times in 47 sequences of the guard split, all within one article, and 0 times in the consequence split. The untestable features' median enrichment is 52.6 against 55.4 for the testable ones, so the exclusion does not select on the guard score. CAR@K at $K = 10, 25, 50, 100, 150$ is $0.55, 0.55, 0.52, 0.45, 0.45$ (Sec. 2).

Table 12: The positive control on X2's consequence test, per arm. For each selected feature the gradient of its concept token's logit, estimated on the consequence split, is ablated. "Defined": testable features for which the gradient could be computed (204 on the two 160m arms, a different set from the 204 paired features of Table 3). "Detected": the rate at which H supports the ablated direction ($S_f > 1.96$). "Wrong sign": the rate at which ablating it damages prediction of the concept *less* than prediction elsewhere, so the control's own manipulation did not have its intended effect. The control is used on the two arms where that rate is below 0.10. Validity and detection use the same score, so the detection rate on those arms is conditional on that choice. Rows are generated from the result file.

arm	defined	detected	wrong sign
Pythia-160m L4	94	0.79	0.00
Pythia-160m L8	110	1.00	0.00
Pythia-410m L16	102	0.55	0.19
Pythia-410m L8	99	0.07	0.93

A second intervention. Injecting the feature where it is silent and asking whether the concept is helped is a different claim from removing what it contributes. Activation supports 0.65 of the 405 paired features: 162 of the 183 that ablation supports, and 100 of the 222 it does not. Part of the disagreement is therefore specific to ablation: counting either intervention as support gives 0.70, and 0.30 of the paired features are supported by neither. Activation was not the prespecified endpoint and has no matched controls, so it is reported beside CAR.

X1 and X3 per configuration. Of the 5 nulls in each X1 configuration, in the order listed above, R^2 alone accepts $0, 0, 1, 1$ and CKA alone accepts $4, 0, 4, 4$. For X3 the counts are $0, 0, 0$ and $4, 4, 4$.

Reproduction. `xai_cka.py`, `xai_esm.py` and `xai_sae.py` in the release run the three experiments end to end and write the tables above as JSON.

G Where the four operations came from

The four operations were chosen from an annotation of **1,547 prize events** (2,327 award records across 23 prizes, 1731–2026) by what the contribution did: it changed the **facts** held, the **instruments** available, or the **description** used. Changes of description form the ladder of Fig. 2. Across all fields 45% of events change a description. The physics subset (532 events) has nine physics prizes (the Nobel, Wolf, Dirac, Sakurai, Onsager, Heineman, Boltzmann, Lorentz and Breakthrough Fundamental Physics prizes) plus the Kavli Prize, whose fields include neuroscience and nanoscience. In it 65% of events change a description, and 69% with the Max Planck Medal in place of the Kavli Prize. This is why the operations are built for physics. Sec. 3 builds operations for four of the six rungs; L0 is what automated discovery already does, and L5 was not attempted. The corpus has a single annotator, a language model reading each official citation under human direction, and it supports no claim in this paper (Sec. 9).

Table 13 places fourteen automated-discovery systems on the ladder by the operation each publishes, which is a statement about published operations and leaves open what each could be extended to do. They cluster on L0; DreamCoder reaches L2 by naming reusable structure, and predicate invention aims at L4. We found none that performs L1: none takes a theory that already fits and returns one that fits identically while carrying less.

H External replication: protocol details

Tolerance. On the labelled suite of Table 14, whose systems are accepted at every separation tested when untilted (a different construction from the system of Table 5, whose floor lies between $12\times$ and $25\times$), tilting the true subspace toward the fast one makes the guard pair refuse it from 0.05 rad at $62.9\times$, 0.08 rad at $25.2\times$ and $12.6\times$, and 0.03 rad at $6.3\times$ separation. At strong separation a small tilt already mixes in modes that are much faster; at weak separation closure sits near its threshold before any tilt.

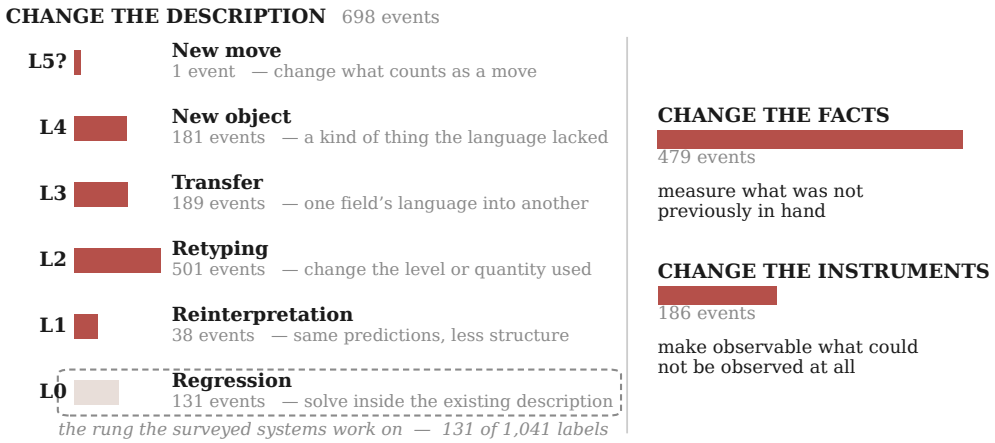


Figure 2: What a discovering system has to be able to do. A move changes the facts you hold, the instruments you have, or the description you use; only a change of description has a depth, and its six rungs are **regression** (L0), solving inside the description you already have; **reinterpretation** (L1), the same predictions with less structure; **retyping** (L2), changing the level or quantity it is written in; **transfer** (L3), carrying another field's language in; **new object** (L4), adding a kind of thing the language lacked; and **new move** (L5), changing what counts as a move. The three column totals count annotated prize events across all fields (1,363 of the 1,547; the other 184 fall in five smaller classes). The per-rung counts are depth labels, since an event may sit on more than one rung (1,041 labels on the 727 events that carry one). Bar lengths compare within a column: a square-root scale on the ladder, a linear one on the right. The fourteen systems we surveyed operate within a predefined representational or operator language; L5 is a single event its annotator flagged uncertain.

What was skipped and why. Of 137 systems in `dysts`, 129 ran. Eight were skipped: two abstract base classes with no concrete right-hand side, and six delay-differential systems (`IkedaDelay`, `MackeyGlass`, `PiecewiseCircuit`, `ScrollDelay`, `SprottDelay`, `VossDelay`) whose state is a function on an interval rather than a vector in $\mathbb{R}^d$, so a finite-dimensional linear reduction is not defined for them. No system was dropped after seeing its result.

Jacobians. The proposer needs a linear operator. Where `dysts` ships an analytic Jacobian we use it; where it does not, we take central differences of the published right-hand side, which is the same operator. Systems are not excluded for lacking an analytic Jacobian.

The five systems where the constant reduction does not close are `BlinkingRotlet`, `BlinkingVortex`, `InteriorSquirmer`, `SwingingAtwood` and `TurchinHanski`. These are numerical failures of evaluation, and the closure condition itself holds for them. The constant reduction lifts its coarse state to the fine-state origin, and there the published right-hand side of each is undefined: the first two return NaN and the other three divide by zero. Both the derived and the learned coarse law then have a non-finite right-hand side, the closure error is recorded as non-finite, and a non-finite error counts as not closing. A law fitted to finite differences of the coarse trajectory would have zero targets and would close. With the origin moved off the attractor all five close with error 0.000, and the zero-dimensional reduction, which evaluates no right-hand side, closes on all 129. An earlier version of this appendix attributed the five to time dependence and a non-invariant origin; that explanation was wrong, since a constant coarse trajectory satisfies $\dot{q} = 0$ whatever the fine dynamics. The coordinate check runs on every system in `dysts`; the counts here are over the 129 of the main sweep, and it ships as `external_coordinates.json`. Moving the origin makes the right-hand side of three other systems (`WindmiReduced`, `YuWang`, `YuWang2`) non-finite at the displaced point, which is why the lifted version closes on 126 there rather than on all.

The negative does not depend on how hard we compressed. Keeping half the directions is an aggressive cut, so we repeated the entire sweep at $k = d - 1$, dropping a single direction, the gentlest reduction that is still a reduction, with the same learned-only rule. The real candidate then closes on **1**

Table 13: Fourteen automated-discovery systems placed on the ladder, by the operation each exposes. Most cluster on L0; **we found none that performs L1**.

system	what it does	rung	why not higher
BACON [Langley, 1981]	rediscovers laws from tables	L0	searches a fixed term language
Eureqa [Schmidt and Lipson, 2009]	symbolic regression on motion	L0	operator set given in advance
AI Feynman [Udrescu and Tegmark, 2020]	physics-informed symbolic regr.	L0	recovers known forms; invents no term
PySR [Cranmer, 2023]	evolutionary symbolic regression	L0	fixed primitive set
LaSR [Grayeli et al., 2024]	symbolic regr. with learned library	L0/L2	abstractions are shortcuts, not retypings
DreamCoder [Ellis et al., 2021]	wake–sleep library learning	**L2**	names structure; no theory to reinterpret
FunSearch [Romera-Paredes et al., 2024]	LLM program search + evaluator	L0	searches a fixed language
AlphaGeometry [Trinh et al., 2024]	olympiad geometry	L0	derives inside a fixed axiom set
Robot Scientist [King et al., 2004]	hypotheses with a robot	L0 + facts	chooses experiments; ontology fixed
Coscientist [Boiko et al., 2023]	autonomous chemistry	instruments	extends reach; no description changes
AI Scientist [Lu et al., 2026]	end-to-end ML papers	L0	proposes and tests ideas; no description changes
AlphaFold [Jumper et al., 2021]	protein structure prediction	tractability	makes computable; does not reframe
Cascade-Correlation [Fahlman and Lebiere, 1990]	grows hidden units as needed	L0	capacity, not concepts
predicate invention [Cropper and Dumančić, 2022]	invents relations in logic programs	**L4***	*open after three decades of ILP

Table 14: The L2 guards on a suite whose correct answer is fixed by construction. **Handed the true subspace the guard accepts every one; handed the proposer's estimate of it, it accepts** 0.71 **of them.** The gap comes from the proposer. Meanwhile the closure test alone admits nearly two thirds of the empty candidates, and the pair admits none.

candidate	n	accepted by the pair	
true subspace, known by construction	24	**24 / 24**	recall 1.00
the proposer's estimate of it	24	**17 / 24**	recall 0.71
fast, random, constant	72	**0 / 72**	precision 1.00
closure test alone	72	**47 / 72**	NAR = 0.65

of 129 systems and is accepted by both guards on 0; the constant reduction closes on **124** and pays on **0**; and 0 of the 3×129 empty controls survive both guards. A linear subspace of the mean Jacobian is the wrong object for these curved attractors at either cut. An earlier version of this paragraph reported this sweep under the superseded rule and the unweighted pay score; the numbers here replace it.

One control behaves differently here than on our own system, and the difference is informative. On the slow–fast system of Table 5, the *fast*-mode reduction closed perfectly because fast modes decay to zero. On these dissipative chaotic systems it does not close at all, because the fast subspace of the mean Jacobian still carries attractor motion rather than decaying to nothing. The vacuity is therefore carried entirely by the constant control here. Our claim narrows accordingly: *closure alone cannot distinguish a real reduction from a reduction to nothing*, which holds on 96% of an external benchmark, rather than the stronger claim that every empty reduction closes in every setting.

Reproduction. `corpus/mechanism/external_dysts.py` in the release runs the whole sweep, importing the guards from `l2_coarse.py` beside it; its output ships as `external_dysts.json`.

Table 15: The L2 guards on 129 chaotic systems from Gilpin [2021], dimension 3–10, with guards and thresholds imported unchanged and acceptance resting on the learned coarse law. **The closure test accepts a reduction to nothing on 96% of the benchmark. The pay test refuses it on all of it.** The median closure error is over systems where it is finite. Rows are generated from the result file.

reduction	closes	median closure err.	max pay score	**accepted by both**
slow modes (the real candidate)	0 / 129	1.810	+0.107	**0**
CONTROL constant (all to zero)	124 / 129	0.000	−0.041	**0**
CONTROL fast modes	0 / 129	1.465	+0.065	**0**
CONTROL random subspace	0 / 129	1.902	−0.012	**0**

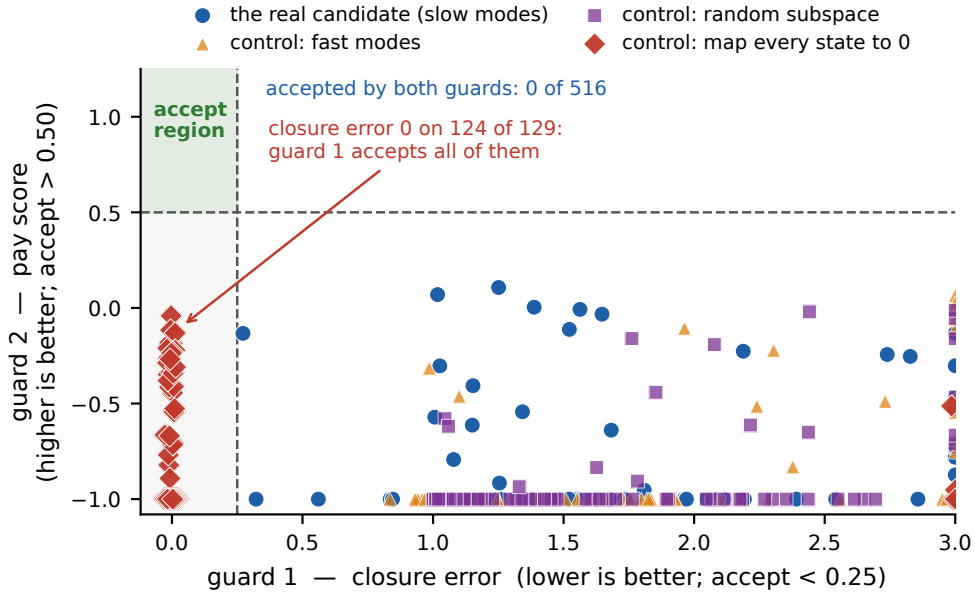


Figure 3: Both L2 guards on 129 chaotic systems from Gilpin [2021]; one marker per (system, candidate reduction), closure scored with the learned coarse law. The red stripe at closure error 0 is the reduction that maps every state to zero: it satisfies guard 1 on 124 of 129 systems and is held out of the accept region by the pay axis, never rising above −0.041. No candidate reaches the accept region. Closure errors that were non-finite or exceeded 3 are drawn at the right edge.

`fig_external.py` regenerates Fig. 3 from that file and *asserts* the two headline counts rather than trusting them, and the release's `verify.py` re-checks both on every run.